

Chapter 12

Surface Area

Unit 4 — Area and Volume

McGraw Hill Glencoe Geometry Texas, 2015

Key Vocabulary

Term	Definition
Polyhedron	3D solid with flat polygon faces
Prism	Polyhedron with two congruent parallel bases
Pyramid	Polyhedron with one base; other faces are triangles meeting at a vertex
Cylinder	Solid with two congruent circular bases
Cone	Solid with one circular base and a vertex
Lateral face	Non-base face of a prism or pyramid
Lateral area (LA)	Total area of lateral faces only
Surface area (SA)	Total area of all faces
Slant height (ℓ)	Height of a lateral face (pyramid or cone)

Euler's Formula

For any **polyhedron**:

$$F + V - E = 2$$

where F = faces, V = vertices, E = edges.

Example: Cube: $F = 6, V = 8, E = 12 \rightarrow 6 + 8 - 12 = 2 \checkmark$

Theorem 12-1: Lateral Area of a Right Prism

$$LA = Ph$$

where P = perimeter of base, h = height.

$$SA = LA + 2B = Ph + 2B$$

where B = area of one base.

Example — Rectangular Prism

Prism: $l = 5$, $w = 3$, $h = 4$

$$LA = 2(5 + 3)(4) = 2(8)(4) = 64$$

$$SA = 64 + 2(5 \times 3) = 64 + 30 = 94$$

Theorem 12-2: Lateral Area of a Right Cylinder

$$LA = 2\pi rh$$

$$SA = 2\pi rh + 2\pi r^2$$

Example: Radius 3, height 7:

$$SA = 2\pi(3)(7) + 2\pi(9) = 42\pi + 18\pi = 60\pi$$

Theorem 12-3: Lateral Area of a Regular Pyramid

$$LA = \frac{1}{2}P\ell$$

where P = perimeter of base, ℓ = slant height.

$$SA = \frac{1}{2}P\ell + B$$

Example — Square Pyramid

Base side = 6, slant height = 5.

$$LA = \frac{1}{2}(24)(5) = 60$$

$$SA = 60 + 36 = 96$$

Theorem 12-4: Lateral Area of a Cone

$$LA = \pi r \ell$$

where r = radius, ℓ = slant height.

$$SA = \pi r \ell + \pi r^2$$

Slant height relationship: $\ell^2 = r^2 + h^2$

Surface Area of a Sphere

$$SA = 4\pi r^2$$

Example: Radius = 6:

$$SA = 4\pi(36) = 144\pi \approx 452.4$$

Surface Area Summary

Solid	Lateral Area	Surface Area
Right Prism	Ph	$Ph + 2B$
Cylinder	$2\pi rh$	$2\pi rh + 2\pi r^2$
Regular Pyramid	$\frac{1}{2}P\ell$	$\frac{1}{2}P\ell + B$
Cone	$\pi r\ell$	$\pi r\ell + \pi r^2$
Sphere	—	$4\pi r^2$

P = perimeter of base, h = height, B = base area, ℓ = slant height

Example — Cone

A cone has radius 5 and height 12. Find the surface area.

Step 1: Find slant height:

$$\ell = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$$

Step 2: Find SA:

$$SA = \pi(5)(13) + \pi(5)^2 = 65\pi + 25\pi = 90\pi \approx 282.7$$

Cross Sections

Solid	Cutting plane	Cross section
Sphere	Any	Circle
Cylinder	$\parallel$ to base	Circle
Cylinder	$\perp$ to base	Rectangle
Cube	$\parallel$ to face	Square
Cube	diagonal	Rectangle
Pyramid	$\parallel$ to base	Similar polygon

Chapter 12 — Formula Summary

Solid	SA Formula
Right Prism	$Ph + 2B$
Cylinder	$2\pi rh + 2\pi r^2$
Regular Pyramid	$\frac{1}{2}P\ell + B$
Cone	$\pi r\ell + \pi r^2$; slant $\ell = \sqrt{r^2 + h^2}$
Sphere	$4\pi r^2$